

Assignment1: CancerMortality

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Introduction

Please refer to the appendix at the end of the report in order to see all the code and outputs produced during the EDA. The first section contains exclusively the explanations and conclusions extracted from the outputs and plots.

This report documents the process and results of the Modeling Assignment for the Statistical Modeling and Inference course. The primary aim of this project is to analyze the Cancer Mortality dataset and build an accurate and interpretable linear model to predict `target_deathrate`, which represents the average cancer mortality rate per 100,000 individuals.

The project begins with data preparation and exploratory analysis, focusing on cleaning and enhancing the dataset to ensure its quality. This involves generating a Data Quality Report, identifying and handling missing values through imputation, and analyzing variable distributions and relationships. Univariate descriptive analysis is performed to examine each variable's characteristics.

The second stage involves modeling using the training dataset, where several key analyses are conducted to optimize the model:

- Initial model exploration addresses potential issues such as multicollinearity and the contribution of each predictor.
- Model transformations are applied to improve variable relationships and address non-linearity.
- An influential data study identifies and evaluates the impact of unusual observations on the model.
- The model is further refined through validation and diagnostics, ensuring it adheres to key assumptions like normality, homoscedasticity, and independence.

Finally, the evaluation process assesses the model's generalization performance on the test dataset, providing insights into its predictive capabilities. A detailed interpretation of the model results is presented, highlighting how the predictors influence the target variable and their practical implications for understanding cancer mortality trends.

This report follows a systematic and methodical approach to ensure the development of a reliable model.

First Steps

To begin, we import the necessary data, load the required libraries, and create functions to streamline repetitive processes.

```
calculate_outliers <- function(data) {
  # Check for regular outliers
  Boxplot(variable, main = "Boxplot")
  outliers <- boxplot.stats(data)$out

  varsumm <- summary(data)
  Q1 <- varsumm[2]
  Q3 <- varsumm[5]
  IQR <- Q3 - Q1

  lower_severe_bound <- Q1 - 3 * IQR
  upper_severe_bound <- Q3 + 3 * IQR
  lower_mild_bound <- Q1 - 1.5 * IQR
  upper_mild_bound <- Q3 + 1.5 * IQR

  # Identify outliers
  mild_outliers <- which((data < lower_mild_bound) | (data > upper_mild_bound))
  severe_outliers <- which((data < lower_severe_bound) | (data > upper_severe_bound))
  num_mild_outliers <- length(mild_outliers)
  num_severe_outliers <- length(severe_outliers)

  # Return the results as a list
  return(list(
    num_outliers = num_mild_outliers,
    num_severe_outliers = num_severe_outliers,
    outliers_indexes = mild_outliers
  ))
}

summary_and_normality_check <- function(data) {
  #Basic summary
  print("Summary: ")
  print(summary(data))

  # Plot normality
  print("Mean")
  mm <- mean(data, na.rm = TRUE); print(mm)
  print("Standard deviation")
  ss <- sd(data, na.rm = TRUE); print(ss)
  hist(data, freq = F, 100, main = "Histogram")
  curve(dnorm(x, mean = mm, sd = ss), col="red", lwd = 2, add=T)

  #Compute skewness
  print("Skewness")
  print(skewness(data, na.rm = TRUE))

  #Number of outliers
  out.res <- calculate_outliers(data)
```

```

print("Num Outliers:")
print(out.res$num_outliers)
print("Num. severe outliers:")
print(out.res$num_severe_outliers)

#Shapiro test
print("Shapiro test: ")
print(shapiro.test(data))

# Most repeated value:
freq_table <- table(variable)
most_repeated_value <- names(freq_table)[which.max(freq_table)]; most_repeated_value
most_repeated_count <- max(freq_table); most_repeated_count
print("Most repeated value ad times it occurs: ")
print(most_repeated_value)
print(most_repeated_count)

return(out.res$outliers_indexes)
}

outliers_df <- df
outliers_df[,] <- 0

```

Explanatory Data Analysis

In this section, we undertake a crucial step in any data science project, particularly for modeling purposes. A detailed descriptive analysis is performed on each variable, highlighting their most relevant characteristics. We follow a column-by-column examination of the dataset, and then generate a data quality report and prepare the data for the subsequent modeling stage by addressing missing values through imputation methods, and deleting non-informative variables and creating new ones.

Note that the explanatory data analysis is assessed using exclusively the train data, even though new variables and variable removals also impact the test dataset.

Univariate Descriptive Analysis

In this initial phase, we analyze all variables in the dataset, highlighting their most significant characteristics. We also convert numeric variables that represent qualitative concepts into factors and consider creating new variables based on existing numeric ones.

Subsequently, we present a data quality report that assesses the overall quality of the data and discusses how it may affect the modeling process.

Variable *avganncount*

avganncount is a numerical variable representing the “Mean number of reported cancer cases diagnosed annually (2010-2015).” First, note that it contains 126 observations with the value “1962.667684”, which we consider inconsistent and that represent NAs. After setting this individuals as NAs, we start the variable analysis.

The difference between the mean (524.24) and the median (155) indicates that the distribution is not symmetric and deviates from normality. Additionally, the first and third quartiles (74 and 383, respectively) are much lower than the mean and the maximum value of 38150, further suggesting skewness. As expected, the Shapiro-Wilk normality test gets rejected. The boxplot confirms this skewness, with an extended upper whisker and numerous outliers, indicating a heavy right tail. It has 207 mild outliers and 131 severe outliers.

```
variable <- df$avganncount
l11 <- which(variable == 1962.667684 ); length(l11)
variable[l11] <- NA
df$avganncount[l11] <- NA
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"avganncount"] <- 1
outliers <- which(outliers_df$avganncount == 1); length(outliers)
```

Variable *avgdeathsperyear*

avgdeathsperyear is a numerical variable representing the “Average number of reported cancer-related mortalities per year.”

This variable is complete, with no missing values. Descriptive statistics reveal a mean of 191.6 and a median of 62, indicating a strongly non-symmetric distribution. This asymmetry is further supported by the first quartile (29) and third quartile (140.5), both significantly lower than the mean, suggesting a right-skewed distribution. Visual exploration using a boxplot confirms this skewness, displaying a pronounced right tail.

Additionally, a histogram with an overlaid normal curve underscores the substantial deviation from normality, a conclusion reinforced by the results of the Shapiro-Wilk test. The distribution exhibits a significant number of outliers, with 225 classified as mild and 138 as severe, all of which are above the third quartile. These

outliers contribute to the variable's extended right tail and may warrant further investigation to understand their context and potential implications.

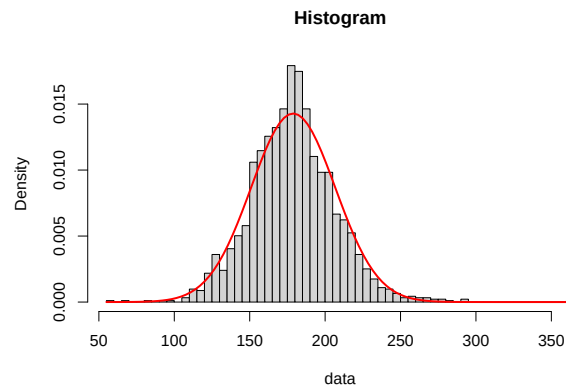
```
df$avgdeathsperyear <- as.numeric(df$avgdeathsperyear)
variable <- df$avgdeathsperyear
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "avgdeathsperyear"] <- 1
```

Variable *target_deathrate* :

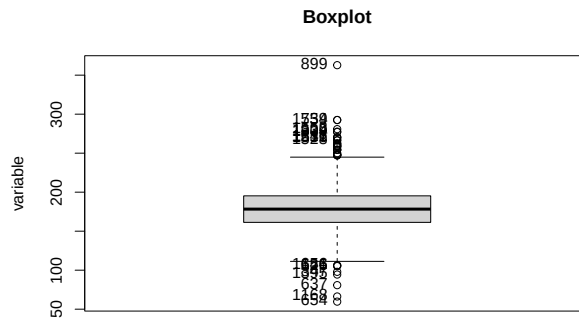
target_deathrate is the numeric target variable of the dataset. We analyze it carefully in the subsequent section **Target variable: Profiling and serial correlation**.

```
variable <- df$target_deathrate;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      59.7  161.3   178.3   178.8   195.3   362.8
## [1] "Mean"
## [1] 178.8197
## [1] "Standard deviation"
## [1] 27.93042
```



```
## [1] "Skewness"
## [1] 0.3243216
```



```
## [1] "Num Outliers:"
## [1] 35
## [1] "Num. severe outliers:"
## [1] 1
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98647, p-value = 4.149e-12
##
## [1] "Most repeated value ad times it occurs: "
## [1] "173.6"
## [1] 8
```

```
outliers_df[res.var,"target_deathrate"] <- 1
```

Variable *incidencerate* :

incidence_rate is a numerical variable representing the “Mean per capita cancer diagnosis rate per 100,000 individuals.”

Anomalies were identified in the data: the same 126 individuals who had a value of “1962.667684” in the *avganncount* variable also showed a value of “453.5494221” in the *incidencerate* variable. These values are considered non-coherent, and thus, we set them as missing (NA). After setting these values to NA, we proceed with the analysis of the column’s distribution.

The mean of the variable is 448.7, the median is 448.7, and the standard deviation is 57.85, indicating moderate variability in the data. The skewness of 1.40 suggests a positively skewed distribution, with the tail extending toward higher values. This is further confirmed by the presence of 36 outliers, including 4 severe outliers, which are all above the upper threshold. The Shapiro-Wilk strongly rejects the null hypothesis of normality.

These results suggest that while the variable is mostly symmetric within its central range, the skewness and presence of outliers may have implications for modeling and analysis.

```
variable <- df$incidence_rate;
ll1 <- which(variable == 453.5494221 ); length(ll1)
variable[ll1] <- NA
```

```
df$incidencerate[111] <- NA
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"incidencerate"] <- 1
```

Variable *medincome* :

medincome is a numerical variable representing the “Median income per county.” The distribution is neither normal nor symmetric, as evidenced by the histogram, Shapiro-Wilk test, and skewness value (1.47), all indicating a right-skewed distribution with a long tail. Descriptive statistics show a minimum of 22,640, a maximum of 122,641 (Loudoun County, Virginia), and a mean of 47,278 with a standard deviation of 12,342. The median income is 45,454, with the first and third quartiles at 39,031 and 52,612, respectively. There are no missing values, but the variable contains 69 outliers, 15 of which are severe, all above the upper quartile. These findings suggest substantial variability and a pronounced concentration of counties with lower incomes compared to a smaller group with exceptionally high incomes.

```
df$medincome <- as.numeric(df$medincome)
variable <- df$medincome;
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"medincome"] <- 1
```

Variable *popest2015* :

popest2015 is a numerical variable representing the population of each county in 2015.

It contains no missing values but has an extensive range and an extremely skewed distribution, as highlighted by a skewness value of 14.71 and a Shapiro-Wilk test confirming strong deviation from normality. The distribution is heavily right-skewed, with a mean of 106,841 significantly exceeding the median of 27,158, and a wide spread between the 1st quartile (12,191) and 3rd quartile (66,880). The variable includes 252 mild outliers and 162 severe outliers, all exceeding the upper whisker of the boxplot. The highest value, 10,170,292, corresponds to Los Angeles County, California. The standard deviation of 376,494 reflects the substantial variability, driven by a small number of extremely populous counties. This skewed distribution highlights the vast disparity in county populations, with most counties having smaller populations and a few extreme outliers significantly influencing the overall statistics.

```
df$popest2015 <- as.numeric(df$popest2015)
variable <- df$popest2015;
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"popest2015"] <- 1
```

Variable *povertypercent* :

povertypercent is a numerical variable representing the percentage of the population in poverty. It has no missing values and a moderate positive skew, as indicated by the skewness value of 0.89 and the Shapiro-Wilk test, confirming the variable does not follow a normal distribution. The mean is 16.79, slightly higher than the median of 15.7, reflecting the slight skew. The 1st quartile (12.15) and 3rd quartile (20.4) show a moderate spread, with values ranging from a minimum of 3.7 to a maximum of 44. The standard deviation is 6.41, indicating variability in poverty levels across counties.

The variable has 42 mild outliers, but no severe ones. The most frequent value is 14.7, occurring 22 times. While this variable shares a non-normal distribution with other numerical variables like *medincome* and *incidencerate*, its skewness and outlier characteristics are less pronounced, indicating a more balanced distribution overall.

```
variable <- df$povertypercent;
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "povertypercent"] <- 1
```

Variable *studypercap* :

studypercap is a numerical variable representing the “Per capita number of cancer-related clinical trials per county.”

A significant feature of this variable is that 1,162 out of 1,831 counties (63.5%) have a value of 0, making this the most frequent value and dominating the distribution. Nearly all non-zero values are flagged as outliers due to their extreme deviation from the majority, with 307 identified as mild outliers, and 222 severe ones.

Given the predominance of zero values and the highly skewed nature, this variable may contribute more noise than predictive value to the model. Removing it from both the training and test datasets is a reasonable decision to simplify the model and enhance its performance.

```
variable <- df$studypercap;
res.var <- summary_and_normality_check(variable)

# Delete from train, test
lltrain <- which(names(df) == "studypercap"); lltrain
lltest <- which(names(test) == "studypercap"); lltest
df <- df[,-lltrain]
test <- test[,-lltest]
```

Variable *binnedi* :

The variable *binnedi* represents median income per capita binned into deciles, categorizing counties into ten groups based on their income distribution. The frequency table shows a relatively balanced distribution across these bins, with counts ranging from 164 to 195 observations per bin. For instance, the smallest bin (37413.8, 40362.7] contains 164 observations, while the largest bin (48021.6, 51046.4] contains 195. The proportion table confirms this balance, with percentages varying narrowly from 8.96% to 10.65%. It contains no missing values.

Given its nature, *binnedi* is likely related to *medincome*, the median income per county, as both variables represent different perspectives of income distribution. While *medincome* is a continuous variable, *binnedi* simplifies analysis by discretizing income into clear categories. This potential relationship highlights the need to evaluate *binnedi* alongside *medincome* to assess redundancy or complementarity in modeling efforts.

```
variable <- df$binnedinc;
summary <- summary(variable); summary
table(variable)
prop.table(table(variable))
barplot(table(variable), main="Bar Plot of Variable", xlab="Categories", ylab="Frequency")
```

Variable *medianage* :

medianage is a numerical variable that expresses the median age of county residents.

Note that initially, the variable contained 18 nonsensical values, ranging from 349.2 to 624, which were considered incoherent. These values were replaced with NAs, leaving the variable with no other missing values. The cleaned data shows a median age ranging from 23.3 to 59.0 years, with a mean of 40.85 and a

standard deviation of 5.14. The skewness is near zero (-0.024), indicating a nearly symmetric distribution. However, the Boxplot and Shapiro-Wilk test for normality suggests the variable is not normally distributed despite its symmetry. The data contains 52 mild outliers (both above the upper bound and below the lower bound), but none are classified as severe. The most frequent value is 38.8, which occurs 29 times.

```
variable <- df$medianage
table(variable)
df[which(df$medianage > 60), c("medianage")] <- NA
summary(df$medianage)
res.var <- summary_and_normality_check(df$medianage)
outliers_df[res.var, "medianage"] <- 1
```

Variable *medianagemale* :

medianagemale represents the median age of male county residents.

All its values are reasonable, ranging from 23.0 to 60.2 years, with no incoherent or missing entries. The variable has a mean of 39.59 years, a median of 39.50 years, and a standard deviation of 5.19, indicating a central tendency and variability. The skewness is slightly positive (0.161), suggesting a slight asymmetry with a longer right tail. However, the Shapiro-Wilk normality test indicates the distribution deviates significantly from normality. The dataset contains 46 mild outliers.

Comparing *medianagemale* with *medianage*, the two variables exhibit similar distributions and characteristics, which is expected as *medianage* includes both genders.

```
variable <- df$medianagemale
res.var <- summary_and_normality_check(variable)
table(variable)
outliers_df[res.var, "medianagemale"] <- 1
```

Variable *medianagefemale* :

medianagefemale represents the median age of female county residents.

Its values range from 23.6 to 58.2 years. All entries are valid, showing no nonsensical or missing values. The average median age is 42.17 years, slightly higher than that of males, with a median of 42.40 years and a standard deviation of 5.20 years. The data is slightly skewed to the left, as indicated by a skewness of -0.206, and the Shapiro-Wilk test confirms a significant deviation from normality. There are 55 mild outliers.

```
variable <- df$medianagefemale
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "medianagefemale"] <- 1
```

Variable *geography* :

The variable *geography* is a categorical variable that specifies the name of the county for each observation. It contains no missing values and serves as a unique identifier for each row, as all values are distinct. Additionally, it includes the name of the state to which each county belongs.

```
variable <- df$geography;
table(variable)
```

Variable *percentmarried*

The variable *percentmarried* represents the percentage of county residents who are married.

It contains no errors, as all values fall within the valid range of 0 to 100. The variable has a minimum of 23.1% and a maximum of 68.0%, with a mean of 51.9% and a median of 52.5%. The standard deviation is 6.74%, indicating moderate variability in marriage rates across counties. The distribution is slightly left-skewed, with a skewness of -0.65, suggesting a higher concentration of values near the upper end of the range. Although the Shapiro-Wilk test confirms that the distribution is not normal, this skewness does not indicate extreme asymmetry. There are 34 outliers, but none are classified as severe. The most frequent value is 52.2%, which appears 22 times.

```
variable <- df$percentmarried
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "percentmarried"] <- 1
```

Variable *pctnohs18_24*

The numerical variable *pctnohs18_24* represents the percentage of county residents aged 18-24 whose highest level of education is less than high school. The values range from 0.5% to 59.1%, with a mean of 18.29% and a median of 17.2%. The distribution of the variable is positively skewed, with a skewness of 0.89, suggesting that a higher proportion of counties have lower values (fewer residents with less than high school education), while some counties exhibit higher percentages. The Shapiro-Wilk test for normality confirms a significant deviation from normality. There are 35 outliers, with 5 classified as severe.

```
variable <- df$pctnohs18_24
table(df$pctnohs18_24)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "pctnohs18_24"] <- 1
```

Variable *ppcths18_24*

The variable *pcths18_24* represents the percentage of county residents aged 18-24 whose highest level of education is a high school diploma.

Note that there is a county with a 0% (Greeley County, Kansas), which potentially is an error.

The values range from 0% to 72.5%, with a mean of 35% and a median of 34.7%. The standard deviation is 8.96%, indicating substantial variability in high school graduation rates across counties. The skewness is 0.22, suggesting a slight positive skew, meaning that most counties have lower percentages of residents with only a high school diploma, while a few counties exhibit higher values. The Shapiro-Wilk test confirms that the distribution is not normal. There are 33 outliers, none of which are severe.

This variable is related to *pctnohs18_24* (the percentage of residents aged 18-24 without a high school diploma), and we can consider creating a factor variable for characterizing this relation.

```
variable <- df$ppcths18_24
ll<- which(df$ppcths18_24 == 0);
res.var <- summary_and_normality_check(df$ppcths18_24)
outliers_df[res.var, "ppcths18_24"] <- 1
table(df$ppcths18_24)
```

Variable *pctsomecol18_24*

The variable *pctsomecol18_24* represents the percentage of county residents aged 18-24 whose highest level of education attained is some college. However, this variable contains 1,376 missing values, which accounts for approximately 75% of the total observations. Given the high proportion of missing data, further steps will be taken to address the missing values, either through imputation methods to obtain valid results or by deleting the variable if it cannot be reasonably imputed.

```
summary(df$pctsomecol18_24)
```

Variable *pctbachdeg18_24*

The variable *pctbachdeg18_24* represents the percentage of county residents aged 18-24 who have attained a bachelor's degree. It ranges from 0% to 51.8%, with a mean of 6.22% and a median of 5.4%. The standard deviation is 4.52%, indicating notable variability across counties. A key feature of this variable is the presence of 73 zero values, meaning that in many counties, no residents aged 18-24 have a bachelor's degree. The skewness is 1.98, reflecting a strongly positive skew, where most counties have relatively low percentages, while a few counties exhibit exceptionally high values. The Shapiro-Wilk test indicates that the normality assumption is rejected. The variable contains 56 mild outliers, and 10 of them classified as severe.

It may also be particularly insightful when combined with related variables, such as *pcths18_24* and *pctsomecol18_24*, to explore the full spectrum of educational attainment.

```
variable <- df$pctbachdeg18_24
table(df$pctbachdeg18_24)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "pctbachdeg18_24"] <- 1
```

Variable *pcths25_over*

The variable *pcths25_over* represents the percentage of county residents aged 25 and over whose highest level of education is a high school diploma.

It ranges from 8.3% to 52.7%, with a mean of 34.73% and a median of 35.3%. The standard deviation is 7.07%. The skewness of -0.37 reflects a slight left skew, suggesting a concentration of values toward the higher end of the range. The Shapiro-Wilk test indicates that the normality assumption is rejected. The variable has 18 outliers, none of which are severe. The most frequent value is 35.6%, occurring 19 times.

```
variable <- df$pcths25_over
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "pcths25_over"] <- 1
```

Variable *pctbachdeg25_over*

The variable *pctbachdeg25_over* captures the percentage of county residents aged 25 and over with a bachelor's degree as their highest level of education.

The distribution is positively skewed, with most counties at the lower end of the range (2.5% to 42.2%), while a few have much higher values. The mean is 13.3%, close to the median of 12.3%, with a standard deviation of 5.47%. Outliers are identified in 59 counties, including 3 severe ones. The variable does not follow a normal distribution.

This variable is essential for understanding higher education attainment in adults and its variability across counties. When analyzed alongside related variables like `pcths25_over`, it can help reveal regional educational trends.

```
variable <- df$pctbachdeg25_over
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "pctbachdeg25_over"] <- 1
```

Variable *pctemployed16_over*

pctemployed16_over represents the percentage of county residents aged 16 and over who are employed. It ranges from 23.9% to 80.1%, with a mean of 54.21% and a median of 54.5%. The data has a standard deviation of 8.22%. Its skewness of -0.34 suggests a slight left skew, meaning there are slightly more counties with higher employment percentages. The Shapiro-Wilk test rejects the normality assumption. There are 11 outliers, none of which are severe.

Additionally, 82 missing values (NA) are present, which require further handling (imputation section).

```
variable<-df$pctemployed16_over
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "pctemployed16_over"] <- 1
```

Variable *pctunemployed16_over*

The variable *pctunemployed16_over* represents the percentage of county residents aged 16 and over who are unemployed. It spans from a minimum of 0.7% to a maximum of 29.4%, with a mean of 7.86% and a median of 7.5%. The standard deviation is 3.46%, indicating moderate variability across counties. The distribution is positively skewed, with a skewness of 0.92, suggesting that while most counties have lower unemployment rates, a few exhibit significantly higher percentages. The Shapiro-Wilk test confirms that the normality assumption is rejected. There are 42 outliers, including 4 severe ones.

This variable, in conjunction with *pctemployed16_over*, provides a comprehensive view of the labor market across counties.

```
variable<-df$pctunemployed16_over
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var, "pctunemployed16_over"] <- 1
```

Variable *pctprivatecoverage*

pctprivatecoverage is a numerical variable that reflects the percentage of county residents with private health coverage. The values span from 23.4% to 89.6%, with a median of 65.2% and a mean of 64.47%. The interquartile range is relatively narrow, from 57.5% (1st quartile) to 72.1% (3rd quartile), indicating that most counties have moderate to high levels of private health coverage. The standard deviation is 10.64%, showing some variability, while the skewness of -0.38 points to a slight left skew, with more counties concentrated at higher percentages. The distribution rejects the normality assumption, as confirmed by the Shapiro-Wilk test.

The dataset identifies 17 outliers, but none are severe. These represent counties with exceptionally low private health coverage rates, offering insights into regional disparities.

This variable holds significant importance due to its potential connection with the target variable, cancer target deathrate. Private health coverage likely plays a crucial role in access to healthcare services, early detection, and treatment of cancer, making it a key factor in understanding regional variations in cancer-related mortality.

```
variable<- df$pctprivatecoverage
table(variable)
res.var <- summary_and_normality_check(variable);
outliers_df[res.var,"pctprivatecoverage"] <- 1
```

Variable *pctprivatecoveragealone*

The variable *pctprivatecoveragealone* represents the percentage of county residents with private health coverage without any public assistance. It ranges from 16.8% to 78.9%, with a mean of 48.65% and a median of 49.0%, indicating that approximately half of the counties have coverage levels close to the central tendency. The interquartile range spans from 41.5% to 55.5%, with a standard deviation of 10.07%. The skewness of 0.02 indicates an almost symmetrical distribution, and the Shapiro-Wilk test suggests the data does not deviate significantly from normality. However, the variable contains 356 missing values (approximately 20% of the dataset), which may require imputation or exclusion strategies. There are 4 outliers, none of which are severe.

This variable is closely related to *pctprivatecoverage*, which captures overall private health coverage, including public assistance.

```
variable<-df$pctprivatecoveragealone
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctprivatecoveragealone"] <- 1
```

Variable *pctempprivcoverage*

pctempprivcoverage represents the percentage of county residents with employee-provided private health coverage.

Its values range from 14.3% to 70.2%, with a mean of 41.29% and a median of 41.1%, indicating a balanced central tendency. The interquartile range spans from 34.6% to 47.7%, with a standard deviation of 9.53%. The skewness is close to zero (0.11), suggesting a nearly symmetrical distribution, while the Shapiro-Wilk test indicates a slight deviation from normality for a confidence level of 99%, but is not rejected in a confidence of 95% (p-value = 0.02861). The variable contains 7 outliers, none of which are severe.

This variable is integral to understanding employment-related health benefits, especially when analyzed alongside *pctprivatecoverage* and *pctprivatecoveragealone*.

```
variable<-df$pctempprivcoverage
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctempprivcoverage"] <- 1
```

Variable *pctpubliccoverage*

The numeric variable *pctpubliccoverage* represents the percentage of county residents with government-provided health coverage, which is not exclusive and may overlap with other types of coverage. It ranges from

11.2% to 62.7%, with a mean of 36.15% and a median of 36.3%, suggesting a balanced distribution around the central tendency. The interquartile range extends from 30.9% to 41.4%, and the standard deviation of 7.84% . The skewness of -0.03 reflects a nearly symmetrical distribution. The Shapiro-Wilk test suggests that the data does not significantly deviate from normality. There are 13 outliers, though none are classified as severe.

```
variable<- df$pctpubliccoverage
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctpubliccoverage"] <- 1
```

Variable *pctpubliccoveragealone*

The variable *pctpubliccoveragealone* represents the percentage of county residents with government-provided health coverage exclusively. The data ranges from 2.6% to 46.6%, with a mean of 19.15% and a median of 18.7%. Most counties fall within an interquartile range of 14.9% to 23%, while the standard deviation of 6.06% indicates moderate variability. The skewness of 0.47 suggests a slight positive skew, with more counties having lower values. The Shapiro-Wilk test indicates that the normality assumption is rejected, and there are 21 outliers, though none are severe.

This variable should complement *pctprivatecoveragealone*, which measures private coverage without public assistance. Together, they should provide a complete view of the reliance on public versus private health coverage. Counties with higher *pctpubliccoveragealone* may reflect areas with greater socioeconomic challenges or fewer employer-provided insurance options.

```
variable<-df$pctpubliccoveragealone
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctpubliccoveragealone"] <- 1
```

Variable *pctwhite*

The variable *pctwhite* represents the percentage of county residents who identify as White. The values range from 12.27% to 99.69%, with a mean of 83.85% and a median of 89.9%. The interquartile range extends from 77.31% to 95.57%, indicating that the majority of counties have a high percentage of White residents. However, the standard deviation of 16.04% highlights notable variability across counties. The skewness of -1.63 indicates a strong left skew, meaning there are a small number of counties with significantly lower percentages of White residents. The Shapiro-Wilk test confirms that the data is not normally distributed. Outliers are present in 97 counties, of which 7 are severe.

This variable is significant for understanding the racial composition of counties and how it may intersect with socioeconomic and health-related outcomes. For instance, its relationship with variables like *pctpubliccoverage* may provide insights into racial disparities in healthcare access or disease prevalence.

```
variable<- df$pctwhite
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctwhite"] <- 1
```

Variable *pctblack*

The variable *pctblack* represents the percentage of county residents who identify as Black. The values range from 0% to 85.95%, with a mean of 9.08% and a median of 2.32%, indicating that most counties have a

relatively small Black population. The interquartile range extends from 0.65% to 10.87%, and the standard deviation of 14.45% highlights substantial variability across counties. The skewness of 2.27 indicates a strong positive skew, meaning a small number of counties have significantly higher percentages of Black residents. The Shapiro-Wilk test confirms that the data is not normally distributed. There are 224 outliers, with 99 classified as severe, and the most frequent value is 0%, occurring in 44 counties.

This variable provides critical insights into the racial diversity of counties and the distribution of Black populations across regions. Comparing *pctblack* with *pctwhite*, *pctpubliccoverage*, or *pctpubliccoveragealone*, can shed light on racial disparities in access to healthcare, insurance coverage, or socioeconomic opportunities. Analyzing its relationship with cancer *target_deathrate* may help identify potential inequities in health outcomes tied to racial demographics.

```
variable<-df$pctblack
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctblack"] <- 1
```

Variable *pctasian*

The variable *pctasian* represents the percentage of county residents who identify as Asian. The values range from 0% to 37.16%, with a mean of 1.27% and a median of 0.55%, indicating that the majority of counties have a very low percentage of Asian residents. The interquartile range extends from 0.26% to 1.25%, with a standard deviation of 2.59%. The skewness of 7.10 indicates a highly positive skew, where a few counties have significantly higher percentages of Asian residents. The Shapiro-Wilk test confirms the data deviates strongly from normality. There are 198 outliers, of which 107 are severe, and the most frequent value is 0%, occurring in 107 counties.

This variable highlights the uneven distribution of Asian populations across counties. The high skew and large number of outliers underscore the concentration of Asian residents in a few counties, likely in urban or metropolitan areas. Analyzing *pctasian* alongside other racial variables like *pctwhite* or *pctblack* and socioeconomic factors such as *pctprivatecoverage* or *pctpubliccoverage* can help explore racial and ethnic disparities in healthcare access, insurance coverage, and health outcomes.

```
variable<-df$pctasian
res.var <- summary_and_normality_check(variable)
hist(variable , 1000 )
outliers_df[res.var,"pctasian"] <- 1
```

Variable *pctotherrace*

The variable *pctotherrace* represents the percentage of county residents identifying as a race other than White, Black, or Asian. It ranges from 0% to 41.93%, with a mean of 2.00% and a median of 0.78%. Most counties have very low values, as reflected in the interquartile range (0.29% to 2.11%) and the strong positive skew (4.99). Outliers are frequent (181 total, 102 severe), and the most common value is 0%, appearing in 79 counties. The Shapiro-Wilk test indicates a significant departure from normality.

```
variable<-df$pctotherrace
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctotherrace"] <- 1
```

Variable *pctmarriedhouseholds*

pctmarriedhouseholds represents the percentage of households that are married.

Values range from 22.99% to 71.40%, with a mean of 51.40% and a median of 51.73%, indicating a fairly balanced distribution. The interquartile range spans from 47.85% to 55.48%, with a standard deviation of 6.45%. The skewness of -0.50 suggests a slight left skew, and the Shapiro-Wilk test indicates non-normality. There are 57 outliers, including 2 severe ones, and the most common value is 50%, occurring in 2 counties.

```
variable<- df$pctmarriedhouseholds
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"pctmarriedhouseholds"] <- 1
```

Variable *birthrate*

The variable *birthrate* represents the number of births relative to the number of women in each county.

Values range from 0 to 21.33, with a mean of 5.60 and a median of 5.36. The standard deviation is 1.93, and the skewness of 1.25 shows a moderate positive skew, with some counties having significantly higher birthrates. The Shapiro-Wilk test confirms non-normality. There are 104 outliers, including 17 severe ones.

```
variable<-df$birthrate
table(variable)
res.var <- summary_and_normality_check(variable)
outliers_df[res.var,"birthrate"] <- 1
```

Definition of factors

In this section, we convert selected numerical variables into categorical variables (factors). This process enhances the dataset by grouping continuous data into meaningful categories, allowing the modeling process to capture patterns more effectively and potentially improving model fit. By creating factors, we enable the model to identify relationships and trends that might be obscured in the original numerical form.

Note that new factor variables are created both in the training and test sets.

medincome factor

We have created a factor for *medincome* by dividing it into quartiles, categorizing counties into “Very Low”, “Low”, “Moderate” and “High” income levels. With this factor we’ll simplify the analysis and also allows us to capture non-linear relationships between income and the target variable.

```
quantiles <- quantile(df$medincome, probs = c(0, 0.25, 0.5, 0.75, 1))
df$f.income <- cut(
  df$medincome,
  breaks = quantiles,
  labels = c("Very Low", "Low", "Moderate", "High"),
  include.lowest = TRUE
)

quantiles <- quantile(test$medincome, probs = c(0, 0.25, 0.5, 0.75, 1))
test$f.income <- cut(
```

```

test$medincome,
breaks = quantiles,
labels = c("Very Low", "Low", "Moderate", "High"),
include.lowest = TRUE
)

```

popest2015 factor

By defining this factor, we seek to divide the counties in four groups regarding its population. This way we can relate variables with the magnitude of each county.

```

quantiles <- quantile(df$popest2015, probs = c(0, 0.25, 0.5, 0.75, 1))
df$f.population <- cut(
  df$popest2015,
  breaks = quantiles,
  labels = c("Very Low", "Low", "Moderate", "High"),
  include.lowest = TRUE
)

quantiles <- quantile(test$popest2015, probs = c(0, 0.25, 0.5, 0.75, 1))
test$f.population <- cut(
  test$popest2015,
  breaks = quantiles,
  labels = c("Very Low", "Low", "Moderate", "High"),
  include.lowest = TRUE
)

```

geography factor

To potentially enhance the model, we propose extracting the state names from geography and creating a new factor representing states. This new factor could capture regional effects and improve the model's ability to identify patterns related to state-level characteristics.

```

df$f.state <- sapply(strsplit(df$geography, " "), function(x) x[2])
table(df$f.state)
df$f.state <- as.factor(df$f.state)

test$f.state <- sapply(strsplit(test$geography, " "), function(x) x[2])
table(test$f.state)
test$f.state <- as.factor(test$f.state)

combined_levels <- union(levels(df$f.state), levels(test$f.state))
df$f.state <- factor(df$f.state, levels = combined_levels)
test$f.state <- factor(test$f.state, levels = combined_levels)

```

education factor 18_24 factor

To create the education level factor we first imputed missing values for the pctsomecol18_24 using its complementary relationship with the other education variables between 18 and 24 years old, ensuring data completeness.

```

miss_ids <- which(is.na(df$pctsomecol18_24))
imp_values <- 100 - df$pctnohs18_24[miss_ids] - df$pcths18_24[miss_ids] - df$pctbachdeg18_24[miss_ids]
variable <- df$pctsomecol18_24
variable[miss_ids] <- imp_values

quantiles_df <- quantile(df$pctsomecol18_24, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_2 <- quantile(variable, probs = seq(0,1,0.1), na.rm = FALSE)
summary(variable)

plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_2, col = "red", lwd = 2);
legend("topright", legend = c("df$pctsomecol18_24", "imp_pctsomecol"),
      col = c("blue", "red"), lwd = 2)

# add imputed values
df$pctsomecol18_24 <- variable

# Repeat the process for the test set
miss_ids <- which(is.na(test$pctsomecol18_24))
imp_values <- 100 - test$pctnohs18_24[miss_ids] - test$pcths18_24[miss_ids] - test$pctbachdeg18_24[miss_ids]
variable <- test$pctsomecol18_24
variable[miss_ids] <- imp_values

quantiles_df <- quantile(test$pctsomecol18_24, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_2 <- quantile(variable, probs = seq(0,1,0.1), na.rm = FALSE)
summary(variable)

# add imputed values
test$pctsomecol18_24 <- variable

```

Then we assigned weights to each education level (No HS = 1, HS = 2, Some College = 3, Bachelor = 4) and calculated a weighted average to represent the overall educational attainment for each county. The results were categorized into “Low Education”, “Moderate Education” and “High Education” levels. This factor will help us capturing the impact of educational attainment on the target variable.

```

weights <- c(1,2,3,4)
education_level <- rowSums(df[,c("pctnohs18_24", "pcths18_24", "pctsomecol18_24", "pctbachdeg18_24")]) *
df$f.education18_24 <- cut(
  education_level,
  breaks = c(0, 1.5, 2.5, 4),
  labels = c("Low Education", "Moderate Education", "High Education")
)
table(df$f.education18_24)

education_level <- rowSums(test[,c("pctnohs18_24", "pcths18_24", "pctsomecol18_24", "pctbachdeg18_24")]) *
test$f.education18_24 <- cut(
  education_level,
  breaks = c(-Inf, 1.5, 2.5, Inf),
  labels = c("Low Education", "Moderate Education", "High Education")
)

```

education factor over-25 factor

The education factor over-25 was created to summarize the educational attainment of county residents aged 25 and over. It is based on the ratio of the percentage of individuals with a bachelor's degree (*pctbachdeg25_over*) to those with only a high school diploma (*pcths25_over*). The factor categorizes counties into three levels: "Mostly High School" for a low prevalence of bachelor's degrees, "Balanced Education" for a similar proportion of both levels, and "Mostly Bachelor's" for counties where bachelor's degree attainment is more common.

```
education_ratio <- df$pctbachdeg25_over / df$pcths25_over

# Categorize based on the ratio
df$f.education25_over <- cut(
  education_ratio,
  breaks = c(-Inf, 0.5, 1.0, Inf),
  labels = c("Mostly High School", "Balanced Education", "Mostly Bachelor's")
)

table(df$f.education25_over)

#Same for test set
education_ratio <- test$pctbachdeg25_over / test$pcths25_over

# Categorize based on the ratio
test$f.education25_over <- cut(
  education_ratio,
  breaks = c(-Inf, 0.5, 1.0, Inf),
  labels = c("Mostly High School", "Balanced Education", "Mostly Bachelor's")
)
```

unemployment

The unemployment factor categorizes counties into three levels based on the percentage of unemployed residents aged 16 and over. Counties are classified as "Low Unemployment" (<5%), "Moderate Unemployment" (5-10%), or "High Unemployment" (>10%).

```
df$f.unemployment <- cut(
  df$pctunemployed16_over,
  breaks = c(-Inf, 5, 10, Inf),
  labels = c("Low Unemployment", "Moderate Unemployment", "High Unemployment"),
  right = FALSE
)

table(df$f.unemployment)

#Same for test set:
test$f.unemployment <- cut(
  test$pctunemployed16_over,
  breaks = c(-Inf, 5, 10, Inf),
  labels = c("Low Unemployment", "Moderate Unemployment", "High Unemployment"),
  right = FALSE
)
```

Data Quality Report

This section provides a comprehensive overview of the dataset’s quality, complementing the prior identification of univariate outliers and missing values with a visual summary to highlight key patterns and anomalies. The data quality report is structured into two parts: an analysis regarding the variables and a review of individual records (stating the number of outliers per individual and identifying multivariate outliers).

Variables

This first part of the data quality report assesses the quality of the data regarding the variables.

Missing Values and Errors. The dataset originally contained missing values in three variables: *pctemployed16_over* (82 missing values), *pctprivatecoveragealone* (356 missing values), and *pctsomecol18_24* (1376 missing values). To address this, imputation methods will be applied to fill these gaps. Specifically, *pctsomecol18_24* values were calculated based on its numerical relationship with complementary variables *pctnohs18_24*, *pcths18_24*, and *pctbachdeg18_24*, ensuring the integrity of the data prior to creating the *education_level* factor.

For *pctemployed16_over* and *pctprivatecoveragealone*, missing values will be imputed using the *imputePCA* method to preserve variable distributions.

Additionally, we found 126 erroneous values in *avganncount* and *incidencerate* across the same observations. These values were replaced with NA to avoid introducing bias, ensuring a more reliable analysis.

All missing and erroneous values will be imputed in a manner that maintains the original characteristics of the dataset, supporting robust and unbiased modeling.

Outliers. Outliers were thoroughly examined to identify variables with the highest counts of mild and severe cases, as these can distort data distribution and impact model performance. As stated in the statement, all outliers are retained.

The table below highlights the top 5 variables ranked by the number of outliers in each category. This analysis aids in determining variables that may require transformation or additional handling to mitigate their influence.

Rank	Mild Outliers Variable	Count	Severe Outliers Variable	Count
1	popest2015	252	popest2015	162
2	avgdeathsperyear	225	avgdeathsperyear	138
3	pctblack	224	avganncount	131
4	avganncount	207	pctasian	107
5	pctasian	198	pctotherrace	102

This ranking reveals that *pctprivatecoveragealone* stands out as having the most outliers in both mild and severe categories. Variables such as *popest2015* and *avgdeathsperyear*, *pctblack*, *avganncount*, and *pctasian* also exhibit significant outlier counts.

Individuals

This second part of the data quality report assesses the number of outliers and missing values per individual and identifies which individuals are multivariate outliers.

Number of missing values across individuals Out of the total observations in the dataset, 1306 have no missing values across any variable. Additionally, 371 observations contain a missing value in one variable, 128 observations have missing values in two variables, 23 observations have missing values in three variables, and 3 individuals have missing values in 4 variables(“Kiowa County, Kansas”, “Meade County, Kansas”, and “Dakota County, Minnesota”).

```
#Number of missing values
na_counts <- rowSums(is.na(df))
table(na_counts)
l1<- which(rowSums(is.na(df))==4); length(l1)
df$geography[l1]
```

Number of outliers across individuals Out of all observations, the dataset contains a distribution of outliers across various categories. Specifically, there are 982 observations without any outliers, 343 observations with one outlier, 146 observations with two outliers, 130 observations with three outliers, 93 observations with four outliers, 68 observations with five outliers, 25 observations with six outliers, 21 observations with seven outliers, 15 observations with eight outliers, and 4 observations with nine and ten outliers. The individuals with more outliers are “Loudoun County, Virginia”, “Williamsburg city, Virginia”, “Santa Clara County, California” ,and “Hamilton County, Indiana”

```
#Number of outliers per individual
outliers_counts <- rowSums(outliers_df)
table(outliers_counts)
l1<- which(rowSums(outliers_df)==10); length(l1)
df$geography[l1]
```

Multivariate outliers The *find_valid_moutlier* function identifies the largest combination of numeric variables without missing values that successfully executes the Moutlier function. By iterating through combinations of variables, starting with the largest possible subsets, this function ensures that the process for detecting multivariate outliers incorporates the maximum amount of information available.

```
find_valid_moutlier <- function(df, quantile = 0.95) {
  # Select numeric columns without NA
  valid_numeric_columns <- which(sapply(df, is.numeric) & sapply(df, function(col) all(!is.na(col))))

  # Track total number of variables
  num_vars <- length(valid_numeric_columns)
  cat("Total numeric variables without NAs:", num_vars, "\n")

  # Start testing combinations from the largest size down to 1
  for (k in num_vars:1) {
    cat("Testing combinations of size", k, "\n")

    # Generate all combinations of size k
    combo_indexes <- combn(valid_numeric_columns, k, simplify = FALSE)
    total_combos <- length(combo_indexes)
    cat("Total combinations of size", k, ":", total_combos, "\n")

    for (i in seq_along(combo_indexes)) {
      combo <- combo_indexes[[i]]
      combo_names <- names(df)[combo]
    }
  }
}
```

```

# Attempt Moutlier function
result <- try({
  res_mout <- Moutlier(df[, combo], quantile = quantile)

  # If successful, return the result immediately
  cat("Valid combination found! Indexes:", paste(combo, collapse = ", "), "\n")
  cat("Variable names:", paste(combo_names, collapse = ", "), "\n")

  multoutliers <- which((res_mout$md > res_mout$cutoff) & (res_mout$rd > res_mout$cutoff))
  cat("Number of multivariate outliers:", length(multoutliers), "\n")

  return(list(res_mout = res_mout, variables = combo_names, multoutliers = multoutliers))
}, silent = TRUE)

# Check for errors and move to the next combination
if (inherits(result, "try-error")) {
  cat("Combination failed.\n")
} else {
  return(result) # Stop at the first valid combination
}
}

# If no valid combination is found
cat("No valid combination found.\n")
return(NULL)
}

```

Since the Moutlier function depends on some stochastic processes, one can get different results when executing it several times. Thus, we comment the results obtained in one single execution, which has been selected carefully to represent the average values.

We found 144 multivariate outliers, with the following IDs: 12, 18, 62, 70, 71, 73, 76, 94, 98, 101, 103, 104, 106, 110, 113, 123, 128, 162, 168, 218, 226, 229, 237, 254, 279, 284, 291, 294, 301, 364, 368, 372, 411, 434, 466, 474, 506, 527, 529, 540, 558, 562, 574, 588, 600, 603, 608, 613, 615, 623, 632, 634, 636, 652, 654, 719, 720, 734, 753, 784, 786, 793, 815, 817, 844, 847, 864, 883, 890, 892, 899, 902, 904, 934, 960, 971, 1016, 1046, 1091, 1093, 1094, 1095, 1096, 1112, 1113, 1120, 1122, 1124, 1135, 1137, 1139, 1144, 1154, 1157, 1160, 1168, 1171, 1180, 1183, 1190, 1223, 1227, 1238, 1243, 1246, 1247, 1388, 1389, 1395, 1398, 1411, 1413, 1420, 1468, 1496, 1522, 1525, 1547, 1559, 1577, 1599, 1607, 1609, 1615, 1617, 1623, 1633, 1636, 1637, 1639, 1641, 1643, 1648, 1669, 1670, 1675, 1679, 1680, 1681, 1689, 1692, 1700, 1736, 1794.

```

result <- find_valid_moutlier(df, quantile = 0.95)
res.out <- result$res_mout;
mult.outliers <- result$multoutliers; length(mult.outliers)

# Plot Multivariate Outliers
dev.off()
plot(res.out$md, res.out$rd)
abline(h=res.out$cutoff, col="red", lty=2)
abline(v=res.out$cutoff, col="red", lty=2)

str(df)

```

Finally, we present the multivariate outlier profiling:

The chi-square test results reveal significant relationships between the multivariate outliers and the categorical variables *f.state*, *f.population*, *f.education25_over*, *f.unemployment*, *f.income*.

Multivariate outliers are more frequently associated with very low population sizes, higher proportions of bachelor's degrees among adults, and states like Texas, Alaska, New Mexico, and South Dakota. These outliers are also associated with extremes in unemployment levels and income categories, highlighting their distinct demographic and socioeconomic characteristics.

The analysis of quantitative variables underscores the importance of several key factors in distinguishing outliers. The Eta-squared values reveal strong associations between outlier status and variables. As expected, multivariate outliers are highly associated with the number of univariate outliers (*n_uout*). Moreover, they are also globally associated with racial composition variables like *pctotherrace*, *pctasian*, *pctwhite* and population estimates (*popest2015*). Other significant variables include average annual case counts (*avganncount*), poverty percentage, and high school education levels. These variables exhibit notable differences between outlier and non-outlier groups, underscoring their explanatory power.

A closer look at the profiles of quantitative variables by cluster reveals distinct patterns. Multivariate outliers are characterized by higher percentages of Asian and Other(not white, black, asian) race populations, lower percentages of White residents, higher poverty levels, larger average annual counts, and younger median ages.

Statistically significant differences between clusters confirm the importance of the identified variables in distinguishing multivariate outliers. Variables related to population demographics, education levels, and racial composition show the most pronounced separations between clusters.

```
bi_mvout_df <- df
#Binary factor if its a multivariate outlier
bi_mvout_df$mout <- 0
bi_mvout_df[mult.outliers, "mout"] <- 1
bi_mvout_df$mout <- factor(bi_mvout_df$mout, labels=c("Not M.O", "Yes M.O"))
table(bi_mvout_df$mout)
#Number of univariate outlier variables for a given observation
bi_mvout_df$n_uout <- rowSums(outliers_df)
#Number of missing values in variables for a given observation
bi_mvout_df$n_miss <- rowSums(is.na(df))

summary(bi_mvout_df[, c("mout", "n_uout", "n_miss")]); sd(bi_mvout_df$n_uout); sd(bi_mvout_df$n_miss)
num_vars <- bi_mvout_df[sapply(bi_mvout_df, is.numeric)]
corrplot(cor(num_vars))

catdes(bi_mvout_df, num.var=39, proba=0.05)
```

Imputation

In this section, all the missing values previously identified are addressed through an imputation process. To avoid repetitive code, a function is first defined to perform the imputation consistently across variables. The imputation is then carried out for each variable with missing values, and a brief commentary is provided for each to discuss the imputation approach and its impact on the data.

Note that values are imputed both in the training and test sets, as stated in the statement of the project.

```
imputation_function <- function(variable) {

  num_vars <- df[sapply(df, is.numeric)]
  miss_ids <- which(is.na(df[[variable]]))
```

```

# Perform PCA-based imputation
res_imp <- imputePCA(num_vars)
imp_values <- res_imp$completeObs[, variable]

# Calculate quantiles for validation
quantiles_df <- quantile(df[[variable]], probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_values, probs = seq(0, 1, 0.1), na.rm = FALSE)

# Plot the quantile comparison
plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = paste("Quantile Comparison for", variable))
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c(variable, paste("imp_", variable, sep = "")),
      col = c("blue", "red"), lwd = 2)

return(list(
  na_indexes = miss_ids,
  imp_values = imp_values[miss_ids]
))
}

imputation_function_test <- function(variable) {

  num_vars <- test[sapply(test, is.numeric)]
  miss_ids <- which(is.na(test[[variable]]))

  # Perform PCA-based imputation
  res_imp <- imputePCA(num_vars)
  imp_values <- res_imp$completeObs[, variable]

  # Calculate quantiles for validation
  quantiles_df <- quantile(test[[variable]], probs = seq(0, 1, 0.1), na.rm = TRUE)
  quantiles_imp <- quantile(imp_values, probs = seq(0, 1, 0.1), na.rm = FALSE)

  # Plot the quantile comparison
  plot(seq(0, 1, 0.1), quantiles_df, type = "l", col = "blue", lwd = 2,
        xlab = "Quantile", ylab = "Value", main = paste("Quantile Comparison for", variable))
  lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
  legend("topright", legend = c(variable, paste("imp_", variable, sep = "")),
        col = c("blue", "red"), lwd = 2)

  return(list(
    na_indexes = miss_ids,
    imp_values = imp_values[miss_ids]
  ))
}

```

pctemployed16_over

The variable *pctemployed16_over* presented 82 missing values, a relatively small number, making it suitable for imputation to preserve the dataset's completeness. Using the `imputePCA` function, the imputation process successfully replicated the original distribution of the variable. This outcome was validated through a quantile comparison plot, where the imputed values (represented by a red line) closely matched the original distribution (blue line). The strong alignment between the two confirms that the imputation method effectively preserved the integrity of the variable's distribution

```
imp.res <- imputation_function("pctemployed16_over")
df[imp.res$na_indexes, "pctemployed16_over"] = imp.res$imp_values

imp.res <- imputation_function_test("pctemployed16_over")
test[imp.res$na_indexes, "pctemployed16_over"] = imp.res$imp_values
```

pctprivatecoveragealone

The variable *pctprivatecoveragealone*, with a relatively high number of missing values (356), has shown substantial improvement after imputation. By employing the `imputePCA` function, the original distribution has been effectively preserved, as evidenced by the close alignment of the imputed and observed data.

```
imp.res <- imputation_function("pctprivatecoveragealone")
df[imp.res$na_indexes, "pctprivatecoveragealone"] = imp.res$imp_values
summary(df)

imp.res <- imputation_function_test("pctprivatecoveragealone")
test[imp.res$na_indexes, "pctprivatecoveragealone"] = imp.res$imp_values
```

incidencerate, *avganncount* and *medianage*

Finally, the imputation process was completed for the two variables where 126 erroneous values were replaced with NA and the *medianage* variable, where 18 erroneous values were similarly addressed. After applying imputation, the resulting values were verified to align closely with the expected distributions, ensuring their reliability. These updated values were subsequently incorporated into the main dataframe. To conclude, a final check using the `summary()` function confirmed that all variables in the dataset are now free of missing values, ensuring data completeness for subsequent analyses.

This is only done in the training set since we can't explore the test dataset, and therefore we can't assume we discovered wrong values in some individuals.

```
imp.res <- imputation_function("incidencerate")
df[imp.res$na_indexes, "incidencerate"] = imp.res$imp_values

imp.res <- imputation_function("avganncount")
df[imp.res$na_indexes, "avganncount"] = imp.res$imp_values

imp.res <- imputation_function("medianage")
df[imp.res$na_indexes, "medianage"] = imp.res$imp_values

summary(df)
summary(test)
```

Target variable: Profiling and correlation analysis.

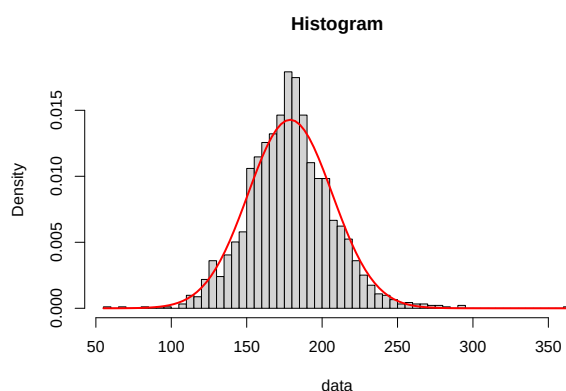
The target variable is complete in the training sample, with no missing values. The median (178.3) and mean (178.8) are nearly identical, suggesting a symmetric distribution. This symmetry is further supported by visual inspections of the boxplot and histogram, both of which confirm the balanced nature of the data around its center.

To evaluate normality, we performed the Shapiro-Wilk test. The resulting small p-value rejected the null hypothesis, indicating that the distribution deviates from normality despite its apparent symmetry. Thus, the target variable is not normally distributed.

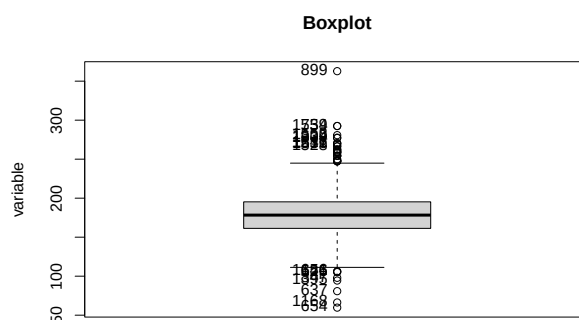
In terms of outliers, the variable contains 35 observations classified as mild outliers, and one severe outlier.

```
summary_and_normality_check(variable)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      59.7  161.3   178.3   178.8   195.3   362.8  
## [1] "Mean"  
## [1] 178.8197  
## [1] "Standard deviation"  
## [1] 27.93042
```



```
## [1] "Skewness"  
## [1] 0.3243216
```



```
## [1] "Num Outliers:"
## [1] 35
## [1] "Num. severe outliers:"
## [1] 1
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98647, p-value = 4.149e-12
##
## [1] "Most repeated value ad times it occurs: "
## [1] "173.6"
## [1] 8

## [1] 43 70 76 339 434 626 637 651 654 720 734 747 843 844 847
## [16] 899 904 1146 1168 1211 1295 1304 1395 1462 1528 1530 1536 1537 1547 1548
## [31] 1558 1559 1570 1639 1656
```

We use the `condes()` function of the `FactoMiner` package in order to profile the numeric target variable.

Using the *quanti* output, we see how the most positively correlated variables with the target are *pctpubliccoveragelone*(0.466), *incidencerate*(0.456) and *povertypercent*(0.450). Moreover, the more negatively correlated are *pctbachdeg25_over* (-0.497), *medincome*(-0.443) and *pctemployed16_over*(-0.440). Moreover, we see that the following numeric variables are not significantly correlated with the target variable, and thus they will be not included in the model: *medianage*, *medianagemale*, *medianagefemale*.

The *quali* output expresses that all factors defined but *feducation18_24* are significantly globally related with the target variable. Thus, this factor won't be included in the model. The most related one is *f.state* ($R^2 = 0.33$), followed by *binnedinc* ($R^2=0.22$) and *f.income* ($R^2=0.18$).

The *category* output shows how some individuals like those who have *f.income*=Very Low, *binnedinc*=[22640, 34218.1], *f.state*=Mississippi, *f.state*=Kentucky or *f.state*=Arkansas have estimates over the overall mean of the target variable. On the other hand, those who have *f.state*=Utah, *f.state*=Arizona, *f.state*=Colorado, *binnedinc*=(61494.5, 125635], among others, have estimates below the overall mean.

```
cor(df[,which(sapply(df, is.numeric))])

res.con <- condес(df[,which(names(df)=="geography"),which(names(df)=="target_deathrate") )
res.con$quanti
non_significant_vars <- setdiff(names(df), c(rownames(res.con$quanti), "target_deathrate")); print(non_
res.con$quali
res.con$category
```

Serial correlations

The ACF plot for *target_deathrate* shows significant autocorrelation for several lags, with a gradual decline over time. The initial lags exceed the confidence intervals, indicating strong serial correlation. In order to assess the serial correlation, the dataframe is randomized and reset. The new dataframe does not include autocorrelation anymore.

```
acf(df$target_deathrate)
ll<- sample(1:nrow(df), nrow(df))
df<-df[ll,]
acf(df$target_deathrate)
```

Modeling

Model Exploration

Base model: Based on the results of the `condes` function executed earlier, the most relevant variables have been selected for inclusion in the model. These variables are as follows:

- Numerical: *pctpubliccoveragealone*, *incidencerate*, *povertypercent*, *pctbachdeg25_over*, *medincome*, *pctemployed16_over*
- Factors: *f.state*, *binnedinc*, *f.income*, *f.education25_over*, *f.unemployment*

As observed in the Residuals vs Fitted plot, the data does not appear to be homoscedastic, indicating non-constant variance in the residuals.

The Q-Q Residuals plot reveals deviations from normality, particularly at the extremes of the X-axis, suggesting the presence of outliers.

In the Residuals vs Leverage plot, there are very few high-leverage points (only one notable case), which is favorable for the model. However, these high-leverage observations may be removed if deemed influential.

To ensure the assumptions of linear modeling are met, appropriate transformations will be applied to the data.

```
m1 <- lm(target_deathrate ~ pctpubliccoveragealone + incidencerate +
         povertypercent+pctbachdeg25_over+medincome+pctemployed16_over+
         f.state + binnedinc + f.income + f.education25_over + f.unemployment,
         data = df)
summary(m1)
plot(m1)
```

Following, we present several plots of the relationship between the target variable and the numeric variables, in order to know if they should be transformed.

By the interpretation of the scatterplots of the target variable vs. the numeric variables *incidencerate*, *povertypercent*, *medincome*, and *pctemployed16_over* variables seem to require a transformation. In the next model, we transform them to avoid

```
#Incidencerate
ggplot(df, aes(x = (incidencerate), y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()

#pctpubliccoveragealone
ggplot(df, aes(x = pctpubliccoveragealone, y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()

#povertypercent
ggplot(df, aes(x = povertypercent, y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()
```

```

#pctbachdeg25_over
ggplot(df, aes(x = pctbachdeg25_over, y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()

#medincome
ggplot(df, aes(x = medincome, y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()

#pctemployed16_over
ggplot(df, aes(x = pctemployed16_over, y = target_deathrate)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal()

```

Apply transformations to variables which shown non-linear relations due to the high skewness of the distribution:

```

m2 <- lm(target_deathrate ~ (pctpubliccoveragealone) + sqrt(incidencerate) +
  log(povertypercent)+ pctbachdeg25_over +log(medincome)+
  log(pctemployed16_over)+ f.state + binnedinc + f.income +
  f.education25_over + f.unemployment, data = df)
summary(m2)
vif(m2)

```

The VIF function allows us to see that there is a lot of multicollinearity introduced by some variables. The variables that are introducing more multicollinearity to the model are: *binnedinc* (GVIF = 3910.12), *f.income* (GVIF=843.77). We remove them from the model

```

m3 <- lm(target_deathrate ~ ((pctpubliccoveragealone) + sqrt(incidencerate)
  + log(povertypercent)+ pctbachdeg25_over +log(medincome)+
  log(pctemployed16_over))* (f.state + f.education25_over
  + f.unemployment), data = df)
summary(m3)

```

We use the step() function in order to see if there is any variable which is not significantly appornting to the model. The obtained model is the following:

```

lm(target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) + pctbachdeg25_over + log(medincome)
+ log(pctemployed16_over) + f.state + f.education25_over + f.unemployment + pctpubliccover-
agealone:f.state + pctpubliccoveragealone:f.education25_over + sqrt(incidencerate):f.state + sqrt(incidencerate):f.education25_
+ pctbachdeg25_over:f.state + log(medincome):f.state + log(medincome):f.education25_over +
log(medincome):f.unemployment + log(pctemployed16_over):f.state + log(pctemployed16_over):f.education25_over
+ log(pctemployed16_over):f.unemployment, data = df)

```

In order to simplify the model further, we applied the Anova function from the car package. After performing the Anova function multiple times, we found that the variables *f.education25_over*, *log(pctemployed16_over)*, *log(medincome):f.unemployment*, *log(pctemployed16_over):f.unemployment*, and *sqrt(incidencerate):f.education25_over* do not significantly contribute to the model. As a result, we decided to remove them to simplify the model.

```

m4 <- step(m3)
summary(m4)

m4<- lm(target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) +
  pctbachdeg25_over + log(medincome) + f.state + f.unemployment + pctpubliccoveragealone:f.state +
  pctpubliccoveragealone:f.education25_over + sqrt(incidencerate):f.state + pctbachdeg25_over:f.state
  log(medincome):f.state + log(medincome):f.education25_over +
  + log(pctemployed16_over):f.state +
  log(pctemployed16_over):f.education25_over , data = df)

summary(m4)
Anova(m4)

```

The residual standard error seems a bit too high. We will now assess a transformation to the target variable in order to try to reduce it while maintaining the R^2 values. First we try a boxcox transformation of the target model. Since the lambda is around 0.8, we apply the boxcox transformation with this value. Thanks to the boxcox transformation, we reduced the residual standard error to 4.633 (which previously was 17.22), while the Adjusted R-squared keeps approximately the same(0.62), while the homoskedasticity has been greatly improved looking at the plots.

```

res_box <- boxcox(target_deathrate ~ pctpubliccoveragealone + sqrt(incidencerate) + pctbachdeg25_over +

index <- which.max(res_box$y)
lambda <- res_box$x[index];lambda

df$target_transformed <- (df$target_deathrate^lambda - 1) / lambda

m5 <- lm(formula = target_transformed ~ pctpubliccoveragealone + sqrt(incidencerate) + pctbachdeg25_over

summary(m5)
par(mfrow=c(2,2))
plot(m5)

```

We still want to reduce the Residual standard error. In order to do it, we apply a log transformation to the transformet target variable, and obtain remarkable good results, reducing the Residual standard error to 0.08, and approximately maintaining the Adjusted R-squared.

```

m6 <- lm(formula = log(target_transformed) ~ pctpubliccoveragealone + sqrt(incidencerate) +
  pctbachdeg25_over + log(medincome) + f.state + f.unemployment + pctpubliccoveragealone:f.state +
  pctpubliccoveragealone:f.education25_over + sqrt(incidencerate):f.state + pctbachdeg25_over:f.state
  log(medincome):f.state + log(medincome):f.education25_over +
  + log(pctemployed16_over):f.state +
  log(pctemployed16_over):f.education25_over ,data = df)
summary(m6)

```

This model has obtained the following results in the training data:

- Residual standard error: 0.08089 on 1535 degrees of freedom
- Multiple R-squared: 0.6757,
- Adjusted R-squared: 0.6134

Bias-variance tradeoff

These models yield excellent results but are susceptible to overfitting the training data. To address this issue, we aim to balance the bias-variance tradeoff by constructing a simpler model. The steps we follow are:

1. Build an initial model using all numeric variables.
2. Address multicollinearity to eliminate insignificant variables.
3. Introduce factors into the model.
4. Use stepwise selection to refine and finalize the model.
5. Incorporate meaningful interactions between numeric variables and factors, ensuring a limited number to minimize the risk of overfitting.

The model obtains the following results in the training data:

- Residual standard error: 18.1 on 1745 degrees of freedom
- Multiple R-squared: 0.5996
- Adjusted R-squared: 0.5801

```
m8 <- lm( target_deathrate ~ avganncount + avgdeathspereyear + sqrt(incidencerate) +
  pctblack + pctwhite+ pctotherrace+ pctasian +pctpubliccoveragealone +
  pctbachdeg25_over + pctnohs18_24 + pctsomcol18_24 + pctbachdeg18_24 +
  pctemployed16_over + pctempprivcoverage + pctmarriedhouseholds +
  f.state + f.unemployment + f.income + f.education25_over+ sqrt(incidencerate):f.unemployment + sqrt
  , data=df)

summary(m8)
plot(m8)
```

Influential data study

Given that the models are being quite negatively affected by influential data, an study of the aforementioned will be performed to address this issue.

Thanks to the influencePlot function, we can see that the following observations (83,1168,1650,80,1389,1591,67,734,168) are potentially influencing strongly in the regression, and thus we want to remove them:

“Queen Anne’s County, Maryland”, “Merrimack County, New Hampshire”, “Tippecanoe County, Indiana”, “Dooley County, Georgia”, “Marion County, West Virginia”, “Grand County, Colorado”, “St. Joseph County, Indiana”, Natrona County, Wyoming”, “Ocean County, New Jersey” .

```
influential <- influencePlot(m8,pch=19,id=list(method="noteworthy",n=3)); rownames(influential)
df$geography[c(899,70,67,734,860,600,168)]
df<- df[-c(899,70,67,734,860,600,168), ]
test[(which(test$f.state=="District of Columbia")),]
test<-test[-562,]
```

Model evaluation

Once the model has been thoroughly studied, this last part evaluates the model in the test data to assess its forecasting capabilities.

We will test Model6 and Model8, to assess the difference between a model carefully built to handle the bias-variance tradeoff and avoid overfitting (Model6), and a model with better results in the training set which was built to get the best results in the training set.

Model 8

Using Model 8, we obtain the following results in the test set: - Mean Squared Error (MSE): 324.9554 - Root Mean Squared Error (RMSE): 18.02652 - SS_{total} is: 918141.9 - SS_{residual} is: 394820.8 - R-squared (R^2): -0.5706384

```
predictions<-predict.lm(
  m8,
  newdata = test,
  interval="prediction",
  level=0.95,
)

inverse_transform <- function(y){
  (lambda*(y)+1)^(1/lambda)
}

predicted <- (predictions[, "fit"])
target_test <- test$target_deathrate

mse <- mean((predicted-target_test)^2); mse
rmse <- sqrt(mean((predicted-target_test)^2)); rmse
mae <- mean(abs(predicted-target_test)); mae
ssres <- sum((target_test-predicted)^2); ssres
sstot <- sum((target_test-mean(target_test))^2); sstot
rsquared <- cor(predicted, target_test)^2; rsquared
r2 <- 1- ssres/sstot; r2
```

Model 6

Using Model 6, we obtain the following results in the test set: - Mean Squared Error (MSE): 0.0583265 - Root Mean Squared Error (RMSE): 0.2415088 - SS_{total} is: 17.62189 - SS_{residual} is: 70.8667 - R-squared (R^2): -3.021516

Note that the negative R-squared shows how the model performs worse than just using a basic model with the average of the test set. The evaluation of this model is very useful to see that obtaining best results in the training data is not always better; the bias-variance trade off must be considered in order to avoid overfitting and end up in a situation in which the model is not able to generalize properly.

```
test$target_transformed <- (test$target_deathrate^lambda - 1) / lambda
test$target_transformed <- log(test$target_transformed)
test$f.state <- factor(test$f.state, levels = levels(df$f.state))

evaluate_model <- function(model, data, target_col, transformed_col = NULL, plot_results = TRUE, print_r) {
  # Generate predictions
  predictions <- predict(model, newdata = data)
  # Handle NAs in predictions
  valid_indices <- !is.na(predictions)
  valid_predictions <- predictions[valid_indices]
  valid_data <- data[valid_indices, ]
  # Determine the column to compare with predictions
  actual_values <- if (!is.null(transformed_col)) valid_data[[transformed_col]] else valid_data[[target_col]]
}
```

```

# Calculate metrics
mse <- mean((valid_predictions - actual_values)^2)
rmse <- sqrt(mse)
ss_total <- sum((actual_values - mean(actual_values))^2)
ss_residual <- sum((actual_values - valid_predictions)^2)
r2 <- 1 - (ss_residual / ss_total)

# Optionally Print metrics
if(print_metrics) {
  cat("Mean Squared Error (MSE):", mse, "\n")
  cat("Root Mean Squared Error (RMSE):", rmse, "\n")
  cat("SStotal is: ", ss_total, "\n")
  cat("SSresidual is: ", ss_residual, "\n")
  cat("R-squared (R²):", r2, "\n")
}

# Optionally plot predictions vs actual values
if (plot_results) {
  # Plot the actual vs predicted values
  plot(actual_values, valid_predictions,
       col = "grey", pch = 19, xlab = "Actual Values",
       ylab = "Predicted Values", main = "Predicted vs. Actual Values")
  abline(a = 0, b = 1, col = "red", lwd = 2.5)
  mean_actual <- mean(actual_values, na.rm=T)
  abline(h=mean_actual, col = "cyan", lwd=2.5)
  # Display SSresidual, SStotal, RMSE, MSE, and R² on the plot
  text(x = max(actual_values) * 0.4, y = max(valid_predictions) * 0.82,
       labels = paste(
         "SS Residual:", round(ss_residual, 2), "\n",
         "SS Total:", round(ss_total, 2), "\n",
         "RMSE:", round(rmse, 2), "\n",
         "MSE:", round(mse, 2), "\n",
         "R²:", round(r2, 2), "\n",
         "Mean Actual:", round(mean_actual, 2))
       , pos = 4, cex = 0.8, col = "green"
  )
  legend("topright", legend = c("Predicted by the model", "Mean of Actual Values"),
       col = c("red", "cyan"), pch = 19, cex = 0.8, box.lwd = 0.5
  )
}

# Return metrics and predictions
return(list( mse = mse, rmse = rmse, r2 = r2, predictions = predictions, ss_residual = ss_residual, ss_total = ss_total))
}

par(mfrow=c(1,1))

results <- evaluate_model(
  model = m6,
  data = test,
  target_col = "target_deathrate",
  transformed_col = "target_transformed",
  plot_results = TRUE
)

```

Conclusion

In conclusion, this project has shown the importance of several foundational aspects of a successful modeling process:

- *Exploratory Data Analysis (EDA)*: The EDA process is fundamental to any data-driven project. It allows us to thoroughly understand the structure and nature of the data, uncover hidden patterns, and identify potential issues such as errors or missing values. EDA is also a key step in creating new, meaningful variables that can improve model performance. Without this critical step, a robust and reliable modeling project cannot exist. It lays the groundwork for all subsequent stages and ensures the data is well-prepared for analysis.
- *Overfitting Awareness*: While achieving good results on the training set can be satisfying, it is essential to remain aware against overfitting. Overfitting occurs when a model becomes too tailored to the training data, capturing noise or insignificant patterns that do not generalize well to unseen data. This can result in poor forecasting and unreliable predictions. It is crucial to remember that the primary objective of modeling is not to optimize performance on the existing training data but to accurately predict new, unseen values. Additionally, overly small mean squared error (MSE) values might signal overfitting and should not be interpreted as the unique indicator of success. Instead, a comprehensive evaluation should include the R2 metric, which measures the proportion of variance in the data that the model captures. A high R2 suggests that the model effectively explains the data's variability, making it a critical measure to consider alongside MSE.